

Suggested Teaching Guideline for Security Concepts PG-DITISS August 2025

Duration: 50 Classrooms hrs + 60 lab hrs (Total: 110 Hrs)

Objective: To introduce the student to issues related to application security and ethical Hacking.

Prerequisites: Knowledge of OS, Networking, programming and database concepts

Evaluation method: Theory exam – 40% weightage
Lab exam – 40% weightage
Internal Assessment– 20% weightage

List of Books / Other training material

Courseware: Gray Hat Hacking: The Ethical Hackers Handbook by Shon Harris

Reference: Python for Unix and Linux System Administration

Note: Every Session includes 2 Hours of theory and 2 Hours of Lab, unless indicated Otherwise.

Application Security

(Web Application Security): (10 hrs Theory+ 14 hrs Lab)

Session 1:

- OWASP Top 10 – 2021
- Injection and Inclusion
- Cross Site Scripting
- Injection in stored procedures

Lab Assignment:

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- Give SQL queries to bypass authentication.
- Make your own cross site script and apply in any web application.

Session 2:

- Denial of Service
- Buffer Overflows and Input Validation
- Access Control

Lab Assignment:

- Try DOS attack to denial a service of any server.
- Try to do input validation to get access in vulnerable application.

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Session 3:

- Web Application Security Risks
- Identifying the Application Security Risks
- Threat Risk Modelling
- Other HTTP fields
- Overview of Burp Suite features

Lab Assignment:

- Identify all risks and vulnerabilities of web applications using tools.
- Configuring Burp Suite with a web browser
- Scanning web applications for vulnerabilities using Burp Suite
- Using Burp Suite's scanner to identify common security issues

Session 4: (2 Theory Hrs + 4 Lab Hrs)

- Data Extraction
- Advanced Identification/Exploitation

Lab Assignment:

- Find vulnerability of data extraction/exploitation of a web application.
- Find vulnerabilities of HTTP Methods (1.0 & 1.1).
- Analyse web application with the help of Wireshark.

Session 5: (2 Theory Hrs + 4 Lab Hrs)

- SAST and DAST tools
- Case Study On Web Application Framework
- Use browser-jsguard Firefox add-on also to detect Malicious and Suspicious Webpages.

Lab Assignment:

- Try to do static and Dynamic application security testing by SAST and DAST tools.

Ethical Hacking (30 hrs Theory+ 36 hrs Lab)

Session 6:

- Security Management Concepts & Principles
- Human side of Information Security's
- Threats of Information System
- Threats and attacks
- Classification of Threats and attacks

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Session 7:

- Protecting Information System Security
- Security in Mobile and Wireless Computing
- Credit card frauds in mobile and wireless Computing
- Information Security Management
- Fundamentals of Information Security

Session 8:

- Cyber Crimes
- Understanding Cyber Crimes in context of Internet
- Legal aspects of open communications
- Indian Penal Law & Cyber Crimes Fraud Hacking Mischief
- International law
- Obscenity and Pornography Internet
- Introduction to Ethical Hacking
- Understanding Ethical Hacking Terminology
- Identifying Different Types of Hacking Technologies
- Understanding the Different Phase Involved in Ethical Hacking

Session 9:

- Types of Hacker Classes
- Red team , Blue Team ,Grey Team
- Ethical Hackers and Crackers
- Goals of Attackers
- Security, Functionality and Ease of Use Triangle
- Defining the Skills Required to become an Ethical Hacker

Session 10: (2 Theory Hrs + 3 Lab Hrs)

- Introduction to Ethical Hacking
- Creating a Security Evaluation Plan
- Types of Ethical Hacks
- Foot-printing and Social Engineering
- Understand How Traceroute Is Used in Foot-printing
- Define the Terms Port Scanning, Network Scanning and Vulnerability Scanning
- Understand various Scanning Methodologies
- SYN, Stealth, XMAS, NULL, IDLE and FIN Scans

Lab Assignment:

- Use NMAP, WHOIS for Reconnaissance.

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- Open-source Intelligence Gathering tool:
 - Netcraft, Shadon, Google Dorks, Recon-Ng
- Use traceroute in footprinting.
- Use FOCA for finger printing.
- Use Tree Pad tool for reporting
- Using Burp Suite for Reconnaissance:
 - Spidering the web application
 - Mapping the attack surface using Burp Suite

Session 11: (2 Theory Hrs + 3 Lab Hrs)

- TCP Communication Flag Types
- Banner Grabbing and OS Fingerprinting Techniques
- How Proxy servers are used in launching an Attack?
- Http tunneling Techniques
- IP Spoofing Techniques
- Enumeration
- Password-cracking Techniques
- Cracking Windows Passwords
- Redirecting the SMB Logon to the attackers
- SMB Redirection, SMB Relay MITM Attacks and Countermeasures
- NetBIOS DOS Attacks
- DDos Attack

Lab Assignment:

- Do all these practical of following;
 - Ip spoofing
 - DNS Reconnaissance
 - DNS Enumeration
 - Host and Port Discovery (using NMAP) etc.
- Use Nessus also to find Vulnerability.

Session 12: (2 Theory Hrs + 3 Lab Hrs)

- Password-Cracking Countermeasures
- Active/Passive online Attacks
- Offline Attacks
- Keyloggers and other Spyware Technologies
- Trojans and Backdoors
- Overt and Covert Channels
- Types of Trojans
- Reverse-connecting Trojans
- Netcat Trojan
- Indications of a Trojan Attacks

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Lab Assignment:

- Use KALI operating system tools to crack password

Session 13:

- Wrapping
- Trojan Construction Kit and Trojan Makers
- The countermeasure Techniques in Preventing Trojans
- Trojan-Evading techniques
- System File Verification

Session 14:

- Difference between a Virus and a Worm
- Types of Viruses
- Antivirus Evasion Techniques
- Virus Detection Methods

Session 15:

- Protocols Susceptible to Sniffing
- Active and Passive Sniffing
- ARP Poisoning
- Ethereal Capture and Display Filters
- MAC Flooding
- DNS Spoofing Techniques
- DNS hacking
- Describe Sniffing Countermeasures

Session 16:

- Types of DOS Attacks
- How DDos Attacks Work
- How BOTs/BOTNETs work
- Smurf Attacks
- SYN Flooding
- Spoofing vs Hijacking
- Types of Session Hijacking
- Steps to perform session Hijacking
- Prevention of session Hijacking

Session 17:

- Hacking Web Servers
- Web Application Vulnerabilities
- Web-Based Password Cracking Techniques
- Wireless Hacking

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- WEP, WPA Authentication Mechanisms and Cracking Techniques
- Wireless Sniffers and Locating SSIDS, MAC spoofing
- Wireless hacking Techniques
- Methods used to secure Wireless Networks

Session 18: (2 Theory Hrs + 3 Lab Hrs)

- Backdoor Devices
- Distributed Dos attacks
- Biometric Spoofing
- Linux Hacking
- Linux Backdoors
- IDSs, Honeypots and Firewalls

Lab Assignment:

- Use NAGIOS, SNORT as IDS/IPS
- Try to install DMZ in your infrastructure.

Session 19: (2 Theory Hrs + 3 Lab Hrs)

- Physical Security
- Overview of Physical Security
- Need of Physical Security
- Factors Affecting Physical Security
- Penetration Testing Methodologies

Lab Assignment:

- Use Metasploit framework for penetration testing.

Session 20: Malware Reverse Engineering (2 Theory Hrs + 3 Lab Hrs)

- Types of Malware
- Malicious code Families
- Latest Trends in Malware
- Analysis of Malware

Lab Assignment:

- Do analysis of a Malware use tools of Static and Dynamic analysis.
- Case Study
- Malware Analysis Scenario – Case Study

Note: We can use **USB Pratirodh** and **AppSamvid** tools for System security.

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Mobile Security: (10 hrs Theory+ 10 hrs Lab)

Session 21:

- Introduction to Android Architecture
- Android File Structure
- Android Build Process
- Android App fundamentals
- Android Security Model
- Device Rooting

Lab Assignment

- Android Emulator setup

Session 22:

- Android Debug bridge
- Penetration Testing Tools

Lab Assignment

- Lab based on Android Debug Bridge

Session 23:

- OWASP Top 10 Mobile App vulnerabilities
- Attacks on Android Apps

Lab Assignment

- Lab based on Reverse Engineering
- Lab based on Smishing attack

Session 24:

- Web based attacks on Android devices
- Networks based attacks
- Social Engineering attacks

Lab Assignment

- Lab based on MITM Attacks
- Packet Sniffing using Wireshark
- Lab based on Phishing Attack

Session 25:

- Overview of Mobile Malware
- Android App Analysis

Lab Assignment

- Static and Dynamic Analysis of Android Apps using MobSF